Practical 2b: Inheritance (All Types)

AIM : To demonstrate all types of inheritance in C#: Single, Multilevel, Hierarchical, and Multiple Inheritance (via Interfaces).

PROGRAM CODE

```
using System;

namespace InheritanceDemo
{
   // ===== SINGLE INHERITANCE =====
   class Animal
   {
      public void Eat() => Console.WriteLine("Animal is eating");
   }

   class Dog : Animal  // Dog inherits Animal
   {
      public void Bark() => Console.WriteLine("Dog is barking");
   }

   // ===== MULTILEVEL INHERITANCE =====
   class GrandParent
   {
      public void Method1() => Console.WriteLine("GrandParent method");
   }

   class Parent : GrandParent
   {
      public void Method2() => Console.WriteLine("Parent method");
   }

   class Child : Parent  // Child -> Parent -> GrandParent
   {
      public void Method3() => Console.WriteLine("Child method");
   }

   // ===== HIERARCHICAL INHERITANCE =====
   class Shape
```

```
{
    public void Draw() => Console.WriteLine("Drawing a shape");
}

class Circle : Shape
{
    public void Info() => Console.WriteLine("This is a Circle");
}

class Rectangle : Shape
{
    public void Info() => Console.WriteLine("This is a Rectangle");
}

// ===== MULTIPLE INHERITANCE via INTERFACES =====
interface IFlyable
{
    void Fly();
}

interface ISwimmable
{
    void Swim();
}

class Duck : IFlyable, ISwimmable
{
    public void Fly() => Console.WriteLine("Duck can fly");
    public void Swim() => Console.WriteLine("Duck can swim");
}

class Program
{
    static void Main(string[] args)
    {
        Console.WriteLine("--- Single Inheritance ---");
        Dog d = new Dog();
        d.Eat();
        d.Bark();
```

```
        Console.WriteLine("--- Multilevel Inheritance ---");
        Child c = new Child();
        c.Method1();
        c.Method2();
        c.Method3();

        Console.WriteLine("--- Hierarchical Inheritance ---");
        Circle cir = new Circle();
        cir.Draw();
        cir.Info();

        Rectangle rec = new Rectangle();
        rec.Draw();
        rec.Info();

        Console.WriteLine("--- Multiple Inheritance (Interface) ---");
        Duck duck = new Duck();
        duck.Fly();
        duck.Swim();

        Console.ReadKey();
    }
  }
}
```

OUTPUT

```
  --- Single
Inheritance ---
Animal is eating
  Dog is barking
  --- Multilevel
Inheritance ---
GrandParent method
  Parent
method
Child
method
  --- Hierarchical
Inheritance ---Drawing a
```